Izmir Institute of Technology

Department of Computer Engineering

CENG543 Critical Summary Report

Fantastically Ordered Prompts and Where to Find Them Overcoming Few-Shot Prompt Order Sensitivity

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1 Introduction and Motivation

The ability of large language models to perform new tasks from only a few examples has positioned in-context learning as a powerful paradigm. Yet this paper brings attention to a critical issue that is often overlooked: extreme sensitivity to the ordering of examples. The authors show that in a four-shot sentiment classification task, certain permutations yield strong performance while others collapse to near-random accuracy (see Appendix Figure A1). This indicates that few-shot learning is far less robust than commonly assumed and depends heavily on a factor that users cannot easily control, namely, the order of the prompt examples. My assessment is that this reveals a deeper limitation in how LLMs generalize from context, since their behavior can be disrupted by simple surface-level rearrangements, challenging theoretical assumptions about the stability of in-context learning. If a model can lose the structure of a task solely due to a change in example order, then the overall reliability of few-shot approaches is severely compromised. The value of the paper emerges here as well: the authors show that this instability persists even in very large models, and the 175B GPT-3 model remains sensitive to ordering on some tasks (see Appendix Figure A2). They also introduce a method to estimate the quality of example permutations without requiring any additional labeled data. This makes the work meaningful both in practice, by improving the stability of few-shot performance, and in theory, by offering new insight into how LLMs process context and why such a seemingly minor factor can produce substantial shifts in model behavior.

2 Overall Framework of the Study and Proposed Method

This work examines the serious example order issue in in context learning and presents a method to address it that does not require additional labeled data. The authors show that different orders of the same few examples can shift performance from nearly SOTA levels to results close to random guessing (see Appendix Figure A1).

To address this problem, an artificial probing set is created using the model's own generative ability. For each example order, the model generates synthetic sentences that resemble inputs from the task (see Appendix Figure A3). The generated labels are discarded because they are unreliable; the evaluation is based only on the model's predictions for these sentences.

The method uses two entropy measures: Global Entropy identifies orders that keep the class distribution of predictions balanced, and Local Entropy measures the model's uncertainty for each sentence to eliminate overconfident orders. The permutations ranked by these scores are filtered to select only the top few. The results show that this selection provides a clear performance improvement across different model sizes and many tasks (see Appendix Table A1).

3 Technical Contributions and Novelty

This study offers a distinct contribution to the in-context learning literature by showing that prompt order sensitivity is not a minor artifact but a core instability that persists across model sizes and tasks. Prior work focused on template quality or example selection, yet the experiments here demonstrate that reordering identical examples can shift performance from near–state-of-the-art to random guessing. In my assessment, this reframes prompt construction as a structural challenge rather than a superficial design choice, especially because performant permutations do not transfer across models and increased sample counts fail to reduce variance. The main novelty of the paper lies in its probing-test approach, which uses the model's own generative output to build an unlabeled substitute development set. Unlike earlier few-shot studies that repurpose part of the already limited examples as validation data to artificially boost results, this method evaluates prompt quality without any held-out labels and reveals the model's genuine few-shot behavior through Global and Local Entropy metrics. In my view, this contributes both conceptually and practically: it exposes a previously hidden failure mode of large language models and introduces a principled mechanism for improving few-shot stability across diverse tasks. Moreover, by contrasting its entropy-based method with a split few-shot setting, such as allocating two examples for training and two for validation, the study adds a useful comparative perspective on why conventional validation-based tuning is insufficient.

4 Experimental Results and Method Evaluation

The study evaluates the proposed entropy-based prompt selection across eleven text classification datasets spanning sentiment, subjectivity, entity classification, and textual entailment (see Appendix Table A2). Four GPT-2 models and two GPT-3 variants are tested under few-shot settings, with 24 permutations and multiple random seeds to measure variance. Accuracy on a 256-sample validation subset serves as the primary metric.

The results show consistent improvements for both Global and Local Entropy, with Global Entropy achieving an average 13% relative gain. These findings strongly support the authors' claim that entropy metrics can

identify high-quality prompt orders without using labeled validation data. The performance gains persist across model scales, including GPT-3 175B, indicating that order sensitivity remains a real limitation even in very large models. The monotonic top-K ranking curves (see Appendix Figure A4) further validate that entropy effectively ranks permutations rather than improving performance by chance.

The paper shows that while the validation-split baseline offers some gains (See Appendix Table A3), it still falls short of entropy-based probing, indicating that the proposed method better reflects true few-shot constraints. However, performance on sentence-pair tasks such as RTE and CB remains low for smaller models, indicating that prompt selection cannot compensate for deeper model weaknesses.

Overall, the experiments are methodologically sound, comparisons appear fair, and the results generalize well across datasets and scales, though benefits are bounded by task complexity and inherent model limitations.

5 Limitations and Critical Discussion

Although the proposed entropy-based probing method provides meaningful gains, the study has several limitations that restrict the generality of its conclusions. First, the approach is evaluated only on text classification tasks, most of which have relatively short and well-structured inputs. It remains unclear whether the same order-sensitivity patterns or the effectiveness of entropy metrics would persist in more complex generative or multi-turn reasoning settings. Additionally, the probing sets are created using the model's own generations, which introduces a strong assumption that these synthetic sentences approximate the true task distribution. In cases where the model produces low-quality or stylistically narrow text, the probing signal may be unreliable.

In terms of experimental design, although the paper includes a supplementary analysis up to 32-shot (Figure 3 - see Appendix Figure A5), the core methodology and entropy-based selection pipeline are evaluated primarily in a 4-shot setting due to context window constraints. This limits visibility into how permutation sensitivity scales in higher-shot scenarios or in prompts containing richer demonstrations. The paper also shows that performant permutations do not transfer across model sizes (see Appendix Figure A4), meaning the method must be rerun for each model variant, which limits its practicality in pipelines with multiple checkpoints. Moreover, evaluating only 24 permutations leaves open whether entropy can scale to realistic few-shot settings, where 8-16 examples yield tens of thousands of possible orders, and raises doubts about its feasibility in real-world pipelines that must score such large spaces. While the comparisons include several baselines, further ablations, such as isolating the effects of sampling temperature or probe length, could clarify whether entropy is the central factor driving improvements.

The results are compelling but not universally strong. For sentence-pair datasets like RTE and CB, improvements remain small for smaller models, suggesting that prompt selection cannot overcome deeper representational limitations. Another unaddressed risk is overfitting to artefacts of the probing set, since both the evaluation and the ranking rely on the same underlying model family.

For future work, extending the evaluation to broader benchmarks, exploring more diverse or structured probing-set generation strategies, and testing robustness across languages, templates, and domains could strengthen the method's generality. In addition, alternative unsupervised indicators, such as diversity-aware sampling, self-consistency signals, or uncertainty-based metrics beyond entropy, may help clarify which properties of the model's behavior truly drive prompt selection performance.

6 Reproducibility and Ethical Considerations

The reproducibility of the study is generally solid, as the authors provide detailed descriptions of prompt construction, permutation generation, entropy-based probing metrics, and evaluation procedures. The experimental setup specifies model sizes, decoding parameters, temperature settings, generation limits, and the use of n-gram blocking, which reduces ambiguity in the sampling process. However, reproducibility remains partially constrained by the reliance on stochastic text generation and implementation-specific decisions that are not fully standardized across frameworks. No implementation code is released, and GPT-3 experiments require access to a paid API, creating barriers for researchers who cannot access identical model versions or inference conditions. Additionally, GPT-3 results are based on a limited number of seeds due to monetary cost, which restricts the ability to replicate variance-related observations.

The study uses widely adopted, publicly available text classification datasets, minimizing data accessibility and privacy concerns. While the method itself does not introduce new ethical risks, it depends on synthetic text generation from pretrained language models, which may embed or amplify model-inherent biases. The paper does not explicitly discuss ethical implications, including concerns about bias propagation or questions of reliability when synthetic probing sets are applied to sensitive domains. Although overall ethical risks are low for the tasks considered, acknowledging potential bias amplification would strengthen the discussion and inform future extensions of the work.

References

[1] Tianyu Gao, Adam Fisch, and Danqi Chen. Making pre-trained language models better few-shot learners. *arXiv preprint*, 2020.

[2] Yao Lu, Marius Bartolo, Adam Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. *arXiv preprint*, 2022.

[3] Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1631–1642, 2013.

Appendix

*Note: All figures in this appendix are taken from the paper "Fantastically Ordered Prompts and Where to Find Them: Overcoming Few-Shot Prompt Order Sensitivity" [**?**].*

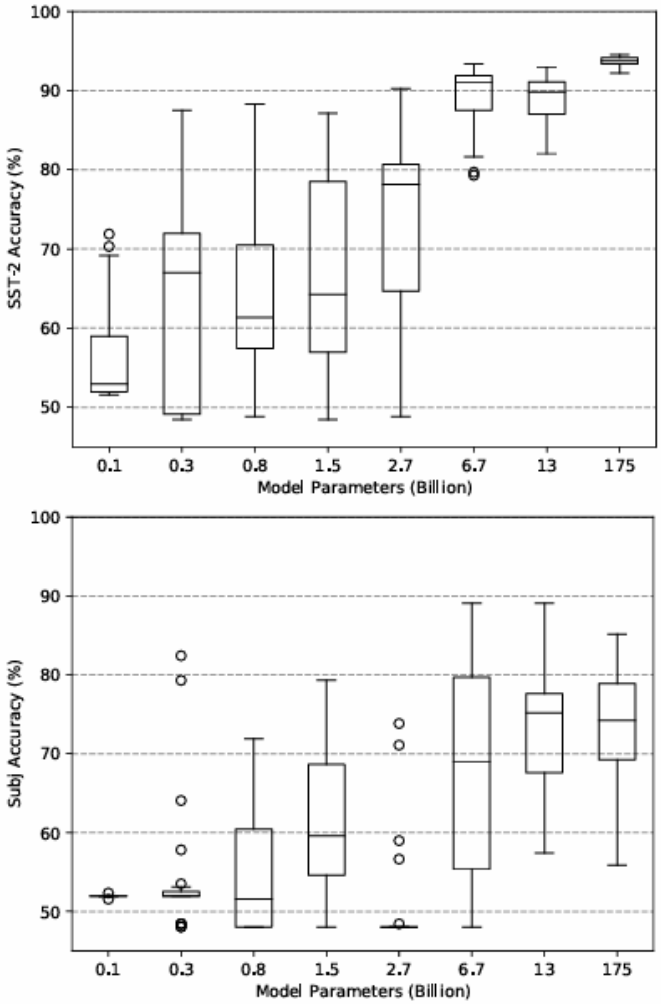

Figure 1: Permutation sensitivity results for the four example sentiment classification task.

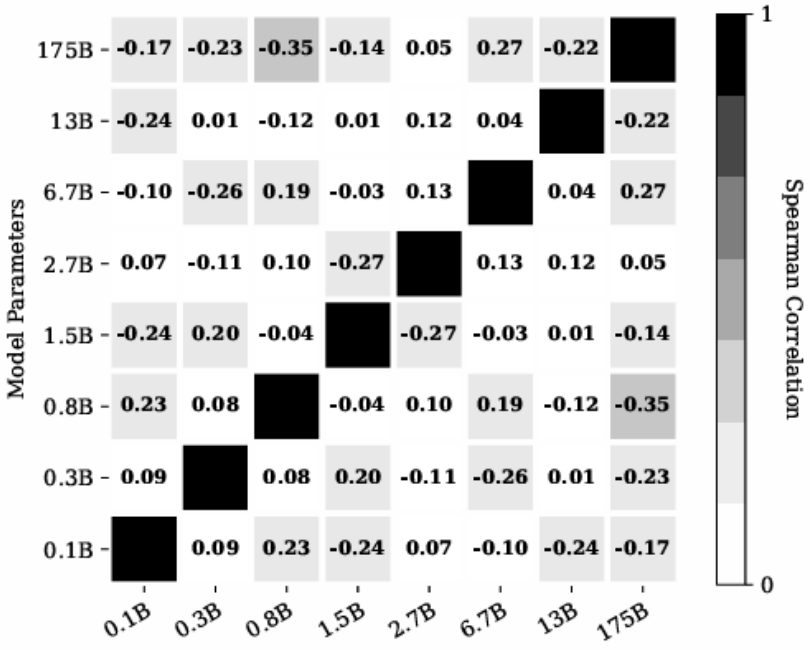

Figure 2: Training sample permutation performance correlation across different models.

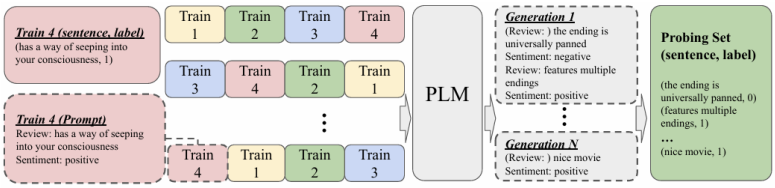

Figure 3: The ordered concatenation of training and test examples turns the in-context learning classification task into a sequence generation task.

	SST-2	SST-5	DBPedia	MR	CR	MPQA	Subj	TREC	AGNews	RTE	CB
Majority	50.9	23.1	9.4	50.0	50.0	50.0	50.0	18.8	25.0	52.7	51.8
Finetuning (Full)	95.0	58.7	99.3	90.8	89.4	87.8	97.0	97.4	94.7	80.9	90.5
GPT-2 0.1B	$58.9_{7.8}$	$29.0_{4.9}$	$44.9_{9.7}$	$58.6_{7.6}$	$58.4_{6.4}$	$68.9_{7.1}$	$52.1_{0.7}$	$\mathbf{49.2}_{4.7}$	$50.8_{11.9}$	$49.7_{2.7}$	$50.1_{1.0}$
LocalE	$\mathbf{65.2}_{3.9}$	$34.4_{3.4}$	$53.3_{4.9}$	$66.0_{6.3}$	$\mathbf{65.0}_{3.4}$	$72.5_{6.0}$	$52.9_{1.3}$	$48.0_{3.9}$	$61.0_{5.9}$	$\mathbf{53.0}_{3.3}$	$49.9_{1.6}$
GlobalE	$63.8_{5.8}$	$\mathbf{35.8}_{2.0}$	$\mathbf{56.1}_{4.3}$	$\mathbf{66.4}_{5.8}$	$64.8_{2.7}$	$\mathbf{73.5}_{4.5}$	$\mathbf{53.0}_{1.3}$	$46.1_{3.7}$	$\mathbf{62.1}_{5.7}$	$\mathbf{53.0}_{3.0}$	$\mathbf{50.3}_{1.6}$
Oracle	$73.5_{1.7}$	$38.2_{4.0}$	$60.5_{4.2}$	$74.3_{4.9}$	$70.8_{4.4}$	$81.3_{2.5}$	$55.2_{1.7}$	$58.1_{4.3}$	$70.3_{2.8}$	$56.8_{2.0}$	$52.1_{1.3}$
GPT-2 0.3B	$61.0_{13.2}$	$25.9_{5.9}$	$51.7_{7.0}$	$54.2_{7.8}$	$56.7_{9.4}$	$54.5_{8.8}$	$54.4_{7.9}$	$52.6_{4.9}$	$47.7_{10.6}$	$48.8_{2.6}$	$50.2_{5.3}$
LocalE	$75.3_{4.6}$	$31.0_{3.4}$	$47.1_{3.7}$	$65.2_{6.6}$	$\mathbf{70.9}_{6.3}$	$67.6_{7.2}$	$\mathbf{66.7}_{9.3}$	$53.0_{3.9}$	$51.2_{7.3}$	$\mathbf{51.8}_{1.0}$	$47.1_{4.2}$
GlobalE	$\mathbf{78.7}_{5.2}$	$\mathbf{31.7}_{5.2}$	$\mathbf{58.3}_{5.4}$	$\mathbf{67.0}_{5.9}$	$70.7_{6.7}$	$\mathbf{68.3}_{6.9}$	$65.8_{10.1}$	$\mathbf{53.3}_{4.6}$	$\mathbf{59.6}_{7.2}$	$51.1_{1.9}$	$\mathbf{50.3}_{3.7}$
Oracle	$85.5_{4.3}$	$40.5_{6.3}$	$65.2_{7.6}$	$74.7_{6.1}$	$80.4_{5.4}$	$77.3_{2.3}$	$79.4_{2.4}$	$63.3_{2.9}$	$68.4_{8.0}$	$53.9_{1.3}$	$62.5_{7.4}$
GPT-2 0.8B	$74.5_{10.3}$	$34.7_{8.2}$	$55.0_{12.5}$	$64.6_{13.1}$	$70.9_{12.7}$	$65.5_{8.7}$	$56.4_{9.1}$	$56.5_{2.7}$	$62.2_{11.6}$	$53.2_{2.0}$	$38.8_{8.5}$
LocalE	$81.1_{5.5}$	$40.3_{4.7}$	$56.7_{7.5}$	$82.6_{4.2}$	$85.4_{3.8}$	$73.6_{4.8}$	$\mathbf{70.4}_{4.2}$	$56.2_{1.7}$	$62.7_{8.1}$	$53.3_{1.6}$	$38.4_{5.2}$
GlobalE	$\mathbf{84.8}_{4.1}$	$\mathbf{46.9}_{1.1}$	$\mathbf{67.7}_{3.6}$	$\mathbf{84.3}_{2.9}$	$\mathbf{86.7}_{2.5}$	$\mathbf{75.8}_{3.1}$	$68.0_{6.5}$	$\mathbf{57.2}_{2.3}$	$\mathbf{70.7}_{3.6}$	$\mathbf{53.5}_{1.5}$	$\mathbf{41.2}_{4.5}$
Oracle	$88.9_{1.8}$	$48.4_{0.7}$	$72.3_{3.3}$	$87.5_{1.1}$	$89.9_{0.9}$	$80.3_{4.9}$	$76.6_{4.1}$	$62.1_{1.5}$	$78.1_{1.3}$	$57.3_{1.0}$	$53.2_{5.3}$
GPT-2 1.5B	$66.8_{10.8}$	$41.7_{6.7}$	$82.6_{2.5}$	$59.1_{11.9}$	$56.9_{9.0}$	$73.9_{8.6}$	$59.7_{10.4}$	$53.1_{3.3}$	$77.6_{7.3}$	$55.0_{1.4}$	$53.8_{4.7}$
LocalE	$76.7_{8.2}$	$\mathbf{45.1}_{3.1}$	$83.8_{1.7}$	$78.1_{5.6}$	$71.8_{8.0}$	$78.5_{3.6}$	$69.7_{5.8}$	$53.6_{3.1}$	$79.3_{3.7}$	$\mathbf{56.8}_{1.1}$	$52.6_{3.9}$
GlobalE	$\mathbf{81.8}_{3.9}$	$43.5_{4.5}$	$\mathbf{83.9}_{1.8}$	$\mathbf{77.9}_{5.7}$	$\mathbf{73.4}_{6.0}$	$\mathbf{81.4}_{2.1}$	$\mathbf{70.9}_{6.0}$	$\mathbf{55.5}_{3.0}$	$\mathbf{83.9}_{1.2}$	$56.3_{1.2}$	$\mathbf{55.1}_{4.6}$
Oracle	$86.1_{1.5}$	$50.9_{1.0}$	$87.3_{1.5}$	$84.0_{2.7}$	$80.3_{3.3}$	$85.1_{1.4}$	$79.9_{5.7}$	$59.0_{2.3}$	$86.1_{0.7}$	$58.2_{0.6}$	$63.9_{4.3}$
GPT-3 2.7B	$78.0_{10.7}$	$35.3_{6.9}$	$81.1_{1.8}$	$68.0_{12.9}$	$76.8_{11.7}$	$66.5_{10.3}$	$49.1_{2.9}$	$55.3_{4.4}$	$72.9_{4.8}$	$48.6_{1.9}$	$50.4_{0.7}$
LocalE	$\mathbf{81.0}_{6.0}$	$42.3_{4.7}$	$80.3_{1.7}$	$75.6_{4.1}$	$79.0_{5.5}$	$72.5_{5.8}$	$54.2_{4.2}$	$54.0_{2.6}$	$72.3_{4.6}$	$50.4_{1.9}$	$50.5_{0.8}$
GlobalE	$80.2_{4.2}$	$\mathbf{43.2}_{4.3}$	$\mathbf{81.2}_{0.9}$	$\mathbf{76.1}_{3.8}$	$\mathbf{80.3}_{3.4}$	$\mathbf{73.0}_{4.3}$	$\mathbf{54.3}_{4.0}$	$\mathbf{56.7}_{2.0}$	$\mathbf{78.1}_{1.9}$	$\mathbf{51.3}_{1.8}$	$\mathbf{51.2}_{0.8}$
Oracle	$89.8_{0.7}$	$48.0_{1.1}$	$85.4_{1.6}$	$87.4_{0.9}$	$90.1_{0.7}$	$80.9_{1.4}$	$60.3_{10.3}$	$62.8_{4.2}$	$81.3_{2.9}$	$53.4_{3.1}$	$52.5_{1.4}$
GPT-3 175B	$\mathbf{93.9}_{0.6}$	$54.4_{2.5}$	$95.4_{0.9}$	$\mathbf{94.6}_{0.7}$	$91.0_{1.0}$	$83.2_{1.5}$	$71.2_{7.3}$	$72.1_{2.7}$	$85.1_{1.7}$	$70.8_{2.8}$	$75.1_{5.1}$
LocalE	$93.8_{0.5}$	$\mathbf{56.0}_{1.7}$	$95.5_{0.9}$	$94.5_{0.7}$	$91.3_{0.5}$	$\mathbf{83.3}_{1.7}$	$75.0_{4.6}$	$71.8_{3.2}$	$\mathbf{85.9}_{0.7}$	$\mathbf{71.9}_{1.4}$	$74.6_{4.2}$
GlobalE	$\mathbf{93.9}_{0.6}$	$53.2_{2.1}$	$\mathbf{95.7}_{0.7}$	$\mathbf{94.6}_{0.2}$	$\mathbf{91.7}_{0.4}$	$82.0_{0.8}$	$\mathbf{76.3}_{3.5}$	$\mathbf{73.6}_{2.5}$	$85.7_{1.0}$	$71.8_{1.9}$	$\mathbf{79.9}_{3.3}$
Oracle	$94.7_{0.2}$	58.2	$96.7_{0.2}$	$95.5_{0.2}$	$92.6_{0.4}$	$85.5_{0.8}$	$81.1_{4.9}$	$77.0_{1.2}$	$87.7_{0.6}$	$74.7_{0.4}$	$83.0_{0.9}$

Table 1: The ordered concatenation of training and test examples turns the in-context learning classification task into a sequence generation task.

Dataset	# of Classes	Avg. Len.	Balanced
SST-2 (Socher et al., 2013)	2	12.4	Yes
SST-5 (Socher et al., 2013)	5	23.1	No
MR (Pang and Lee, 2005)	2	25.7	Yes
CR (Hu and Liu, 2004)	2	22.1	Yes
MPQA (Wiebe et al., 2005)	2	3.9	Yes
Subj (Pang and Lee, 2004)	2	28.9	Yes
TREC (Voorhees and Tice, 2000)	6	11.6	No
AGNews (Zhang et al., 2015)	4	53.8	Yes
DBPedia (Zhang et al., 2015)	14	65.5	Yes
CB (De Marneffe et al., 2019)	3	69.7/8.4	No
RTE (Dagan et al., 2005)	2	55.3/11.9	Yes

Table 2: Used Datasets

	GPT-2 0.1B	GPT-2 0.3B	GPT-2 0.8B	GPT-2 1.5B
Baseline	$58.9_{7.8}$	$61.0_{13.2}$	$74.5_{10.3}$	$66.8_{10.8}$
LocalE	$\mathbf{65.2}_{3.9}$	$75.3_{4.6}$	$81.1_{5.5}$	$76.7_{8.2}$
GlobalE	$63.8_{5.8}$	$\mathbf{78.7}_{5.2}$	$\mathbf{84.8}_{4.1}$	$\mathbf{81.8}_{3.9}$
Split Training Set	$62.8_{5.3}$	$64.2_{6.1}$	$75.1_{6.8}$	$71.4_{7.8}$

Table 3: Used Datasets

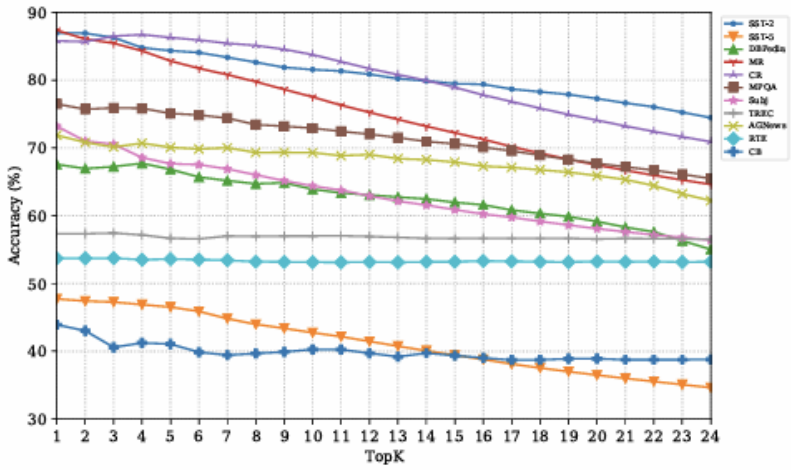

Figure 4: Average performance of different Top K permutation selection on GPT2-Large(0.8B)

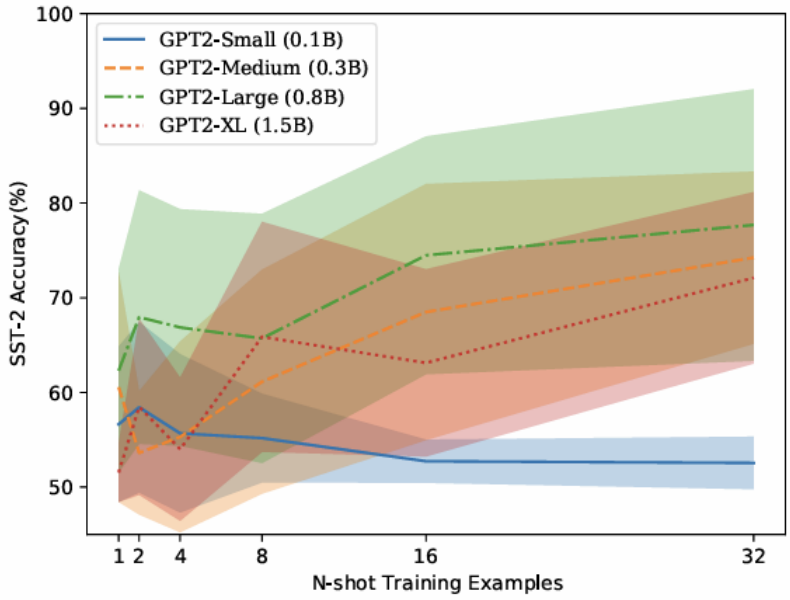

Figure 5: Order sensitivity using different numbers of training samples(Used Dataset - SST -2 [3])